

Verified Generators and Relations for Qupit Clifford Operators

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Bian, Li, Ross, van de Wetering, and Zhao give sixteen rewrite rules that are sound and complete for multi-qudit Clifford circuits over H, S, CZ in every odd prime dimension p . We report a machine-checked proof of the result in safe, axiom-free Agda, against an independently constructed semantics. Rather than follow the paper, which applies one rule set to the whole Clifford group “up to Paulis”, we factor the projective Clifford group as $\overline{\mathcal{P}}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p)$, present the factors separately, and glue the presentations with a generic, verified semidirect-product theorem. The symplectic factor carries the mathematics: the paper’s normal form, refined into a doubly inductive datatype of *normal boxes* and *rows*, is a tower of verified Reidemeister–Schreier coset tables, proved unique by induction on the width, and the paper’s 66 box relations are derived from eighteen natural ones. Verified isomorphisms of presentations reach the paper’s own rules, and a central extension, verified against a cocycle, restores the scalars. Everything is parametric in p and typechecks with no postulates.

CCS Concepts: • **Theory of computation** → **Type theory; Logic and verification; Equational logic and rewriting; Quantum computation theory.**

Additional Key Words and Phrases: Clifford group, qudits, generators and relations, completeness, normal forms, symplectic group, semidirect product, Agda, formal verification

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1 INTRODUCTION

A finite list of circuit equations is *sound* when each holds in the intended semantics and *complete* when every semantic equality between circuits is derivable from it. Completeness is proved by a *normal form*, and the case analysis of every generator against every normal form is enormous: Selinger’s n -qubit Clifford presentation [Selinger 2015] runs a substantial rewrite system through it, its qutrit successor [Li et al. 2025] needs over fifty pages, and its generalisation to all odd primes by Bian, Li, Ross, van de Wetering, and Zhao [Bian et al. 2026a] — sixteen rules for multi-qudit Clifford circuits in every odd prime dimension p , henceforth “the qupit paper” — rests on roughly a hundred pages of derivations [Bian et al. 2026b].

We report a machine-checked proof of that theorem in safe, axiom-free Agda [Norell 2007; The Agda Community 2026], against a semantics constructed independently of the syntax, on the library of [Anonymous 2026a,b] (§4). The proof does *not* follow the paper’s, which works *up to Paulis* with one rule set for the whole group (§3); we instead *factor the group, present the factors, compose the presentations*: $\overline{C}_n = \overline{\mathcal{P}}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p)$ and C_n is a central extension of it by the scalars; the symplectic group gets a coset-table normal form, the Pauli group a compositional presentation, the scalars a cyclic one, generic verified theorems for semidirect products and central extensions glue them, and plumbing reduces the result to the paper’s H, S, CZ and rules.

2 QUPIT CLIFFORD CIRCUITS AND THE MAIN THEOREMS

Fix an odd prime p and $\omega = e^{2\pi i/p}$. On the basis $|0\rangle, \dots, |p-1\rangle$ indexed by $\mathbb{Z}_p$, $X|j\rangle = |j+1\rangle$ and $Z|j\rangle = \omega^j|j\rangle$ generate, with the phases ω^t , the Pauli group $\mathcal{P}_n$, and $H: |j\rangle \mapsto \frac{\delta_p}{\sqrt{p}} \sum_k \omega^{jk}|k\rangle$,

$S: |j\rangle \mapsto \omega^{j(j-1)/2}|j\rangle$, $CZ: |j\rangle|l\rangle \mapsto \omega^{jl}|j\rangle|l\rangle$ generate the Clifford group C_n (δ_p makes the Clifford phases exactly the powers of ω) [Bian et al. 2026a]. Modulo scalars, $\overline{C}_n = C_n / \langle \omega \rangle$ and $\overline{\mathcal{P}}_n \cong \mathbb{Z}_p^{2n}$, $C_n / \mathcal{P}_n \cong \overline{C}_n / \overline{\mathcal{P}}_n \cong \text{Sp}(2n, \mathbb{Z}_p)$, and $\overline{C}_n \cong \overline{\mathcal{P}}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p)$. These “glue facts” are the two not formalised: our semantic groups are *defined* as the right-hand sides.

Circuit $n = \text{Word } (\text{Gen } n)$ is the free monoid on wire-indexed generators over the gates H , S (unary) and CZ (binary), and $_ \uparrow$ shifts a circuit up one wire. The other usual gates are *words* ($Z = HSHHS^{-1}$, $X = HSHHS^{-1}H$, the swap $Ex = (CZ \cdot H \downarrow \cdot H \uparrow)^3$), as is the *multiplier* $M_x |j\rangle = |xj\rangle$, $x \in \mathbb{Z}_p^*$, spelled by the paper over S with a Pauli prefix (its Lemma 2.14) and by us mostly as $R^x H R^{x^{-1}} H R^x H$ with $R = S \cdot Z^{1/2}$. Our primary rule set — Figure 1 modulo scalars, multiplier over R , $(SH)^3 = \varepsilon$ kept as an axiom — is one inductive family $_ \text{CRel}, _ === _$ with sixteen constructors, six one-qubit (orders, the multiplier calculus M-power and semi-MR, comm-HSHHS), seven two-qubit (orders of CZ and Ex , commutations, the swap rules, blake-c12) and three three-qubit (Yang–Baxter, cz-slide, semi-CX $\uparrow$ -CZ $\downarrow$), all drawn in Figure 1, plus the three *structural* rules every circuit theory gets free (congruence under $\uparrow$; gates on disjoint wires commute); $XZ = ZX$ modulo scalars, Ex commuting with CZ and both rules pushing a Pauli through CZ are derived. Figure 1 has fifteen non-scalar rules; under the R -spelling $(SH)^3 = \varepsilon$ must stay an axiom, and §6 verifies the paper’s fifteen in its S -spelling, where it is a theorem.

THEOREM 2.1 (CLIFFORD-PRESENTATION). *For every width n , the sixteen rules present the projective Clifford group:*

Theorem-1: $\forall (n : \mathbb{N}) \rightarrow (n \text{ CRel}, _ === _) \text{ IsPresentationOf } (\text{Pauli} \rtimes \text{Sp } n)$

Theorem-1 = PapPres.presentation

$\text{Pauli} \rtimes \text{Sp } n$ is the semidirect product of $\mathbb{Z}_p^{2n}$ by $\text{Sp}(2n, \mathbb{Z}_p)$ (§6); $_ \text{IsPresentationOf } _$ (§4) is a group isomorphism $\text{Cir } n / \approx \rightarrow \overline{\mathcal{P}}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p) \cong C_n / \langle \omega \rangle$: *soundness*, each rule holds, and *completeness*, $[[_]]$ is injective. Normal forms for symplectic operators (Clifford operators modulo Paulis) are *data*: four kinds of atomic box, the paper’s (its Figure 6), $A = \{(a, b) \in \mathbb{Z}_p^2 \mid (a, b) \neq (0, 0)\}$, $B = D = \mathbb{Z}_p^2$, $E = \mathbb{Z}_p$, spelled as circuits in Figure 2, then rows of boxes and a staircase of rows (§5).

THEOREM 2.2 (UNIQUE-NF). *Normal forms with equal symplectic denotations are equal:*

Theorem-2: $\forall n \{u \ v : \text{NF } n\} \rightarrow [[[u]]] \approx^s [[[v]]] \rightarrow u \equiv v$

Theorem-2 = U.[[]]-injective

3 TWO ROUTES TO COMPLETENESS

The routes share the normal boxes, the symplectic normal form and essentially the same eighteen symplectic relations, and differ in the *organisation* of the argument. The paper’s normal form is a *symplectic normal form* followed by a *Pauli normal form*, unique by Proposition 3.8 (“analogously to the qutrit case”); Section 4.1 pushes the gates in one by one — a gate through a normal box yields an updated box and residual *dirty* gates — collecting the cases into 66 parametrised *box relations* (Theorem 4.4 reduces them to 18, a step already in Agda); relations are applied *up to Paulis*; and Proposition 4.9 boosts by correcting each rule $C_l = C_r$ to $C_l = C_r ; P$ for the unique Pauli P making it projectively sound and replaying symplectic rewrites with the corrected rules, pushing each new Pauli to the end, where it becomes “a potentially different Pauli”. *Why not formalise that*: the procedure is not a function — formalised, it is a coset table, as we write, which would also have to say what happens to the Pauli — and the objects rewritten are circuits modulo Paulis with no syntactic representation of the quotient, so the honest counterpart of Proposition 4.9 is a normaliser on the semidirect product whose well-definedness (congruent inputs give equal cosets and congruent residuals) is exactly what “potentially different Pauli” glosses over; we did not see how to verify it at

reasonable cost. *Our route:* since $\overline{C}_n = \overline{\mathcal{P}}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p)$ and C_n is a central extension of it, we work in a syntax where Paulis are genuinely trivial — symplectic circuits have only H, S, CZ and denote $\text{Sp}(2n, \mathbb{Z}_p)$ itself — and reintroduce Paulis and scalars by construction (Figure 3 in the appendix): $\text{Sp}(2n, \mathbb{Z}_p)$ is presented by a coset tower whose cosets are the normal boxes as a doubly inductive type, whose table is “push a gate through a box” as a structurally recursive function with a verified section, and whose well-definedness is proved *semantically*, by uniqueness one level down; $\overline{\mathcal{P}}_n$ is glued on by the semidirect theorem, whose one new obligation, that the action respects both factors’ rules, follows from soundness and completeness of the factors; plumbing reduces alphabet and relations to the paper’s; and the extension theorem restores the scalars. The Pauli bookkeeping of Proposition 4.9 is thus done once, by a theorem proved for arbitrary groups before this project began.

4 THE FRAMEWORK, IN BRIEF

The library [Anonymous 2026a,b] presents monoids and groups by generators and relations; syntax is unquotiented, so normal forms and coset tables are programs, and every construction comes with a presentation theorem against a concrete group. $\approx$ is the monoid congruence generated on words by the axioms $_ == _$, the word problem of $\langle X \mid \Gamma \rangle$; a Grouplike witness upgrades to the presented group, and $\Gamma \text{ IsPresentationOf } G$ packages it with $[[_]] : \text{Word } X \rightarrow G$ and a group isomorphism from the presented group onto G . A normal form comes with a *section* $\text{inv-nf} : NF \rightarrow \text{Word } X$ realising each normal form as a canonical word; UniqueNormalForm says representatives with equal denotations are equal, and the five-line *by-normalization* turns uniqueness and soundness into completeness, so each case study proves unique, about finite data, never about arbitrary words.

Coset tables. Normal forms follow the chain-of-subgroups recipe of computational group theory [Holt et al. 2005; Sims 1994], the chain for circuits being the wire structure. One level is a Reidemeister–Schreier situation — a subgroup presentation Γ , a group presentation Δ , a finite type C of coset labels with identity I , a section $[_] : C \rightarrow \text{Word } Y$, the embedding f , and the *coset table* $h : C \rightarrow Y \rightarrow \text{Word } X \times C$, pushing one letter past a representative and returning the subgroup residual and the new coset — and the library *verifies* a claimed table against five hypotheses (Appendix B): the table inverts the embedding at I , it respects Δ ’s axioms coset by coset, f respects Γ ’s, $[I] \approx \varepsilon$, and the *soundness law* $[c] \bullet [b]^w \approx (f^w) b' \bullet [c']$ for $(b', c') = h \ c \ b$; from these it derives $\text{NormalForm } \Gamma \text{ NF} \rightarrow \text{NormalForm } \Delta \text{ (NF} \times C)$, and CosetTower iterates up an $\mathbb{N}$ -indexed family, one level per wire. For semidirect products the mixed relation $\text{ConjRel}^w \text{ conj}$ says $h \bullet n = \text{conj } h \ n \bullet h$ with a *word*-valued action, and the semidirect theorem takes the factors’ relation sets, the action, two well-definedness conditions (it respects the acting group’s axioms in its first argument and the normal subgroup’s in its second) and presentations of the factors, and returns a presentation of the semidirect product of the *semantic* groups; extension-presentation (Handbook [Holt et al. 2005], Proposition 2.55) also corrects each relator of the quotient by a word of the kernel, and is proved for central extensions built from a cocycle. The library’s warm-ups — S_n by a tower of swap staircases, all hypotheses but the soundness law closing by evaluation, and $\mathbb{Z}_N \wr S_n$ by the semidirect theorem with finite case splits — are the two patterns the Clifford group combines.

5 THE SYMPLECTIC FACTOR

$\text{Sp}(2n, \mathbb{Z}_p)$ is modelled as the group of $\mathbb{Z}_p$ -linear, form-preserving bijections of the phase space Pauli $n = \text{Vec } (\mathbb{Z}_p \times \mathbb{Z}_p)$ n (the exponents of $Z^a X^b$ up to phase, per wire), with extensional equality, so that the monoid laws hold by refl; the generators act as the paper’s conjugation tables (Lemmas 2.22–2.23), and soundness of every axiom is one clause per axiom. The paper’s symplectic

normal form is a Z -normal then an X -normal layer on n wires, then a normal form on $n - 1$; we refine it into a doubly inductive datatype (Appendix B). An M -row is a column of D boxes ending in an E box (the X -normal layer), an L -row a column of B boxes ending in an A box (the Z -normal layer), and an ML box at width $m \geq 2$ is *either* a pair of rows *or* a bottom D box followed by an ML box one wire up: a spine of D boxes capped by two rows. ML makes explicit the position of the paper's A box and, crucially, makes the coset type at width m a *sum* of the coset type at width $m - 1$ and a flat case, the shape the table and the uniqueness proof recurse on.

The table. The coset table is the paper's "push a gate through a normal box" as a function $\text{ract} : ML(1+n) \rightarrow \text{Gen}(1+n) \rightarrow \text{Circuit } n \times ML(1+n)$ (5 600 lines), with soundness law ract-sound , a chain of box relations and structural rules per clause. The difference from S_n is the hypothesis that the table respects the axioms, which does *not* close by evaluation: its *coset half* is semantic — apply $[[_]]$ to ract-sound on both sides of an axiom, note that residuals live one wire up and cannot move the bottom wire, and conclude by head-injectivity of the coset section (below) — and its *residual half* follows by cancellation, given that $\uparrow$ is injective on congruence classes one width down, a consequence of completeness there. So the tower is a well-founded induction on the width.

Uniqueness. Theorem 2.2 is a structural induction on the staircase from two lemmas: *tail surjectivity*, free since a symplectic transformation carries its inverse, and *head injectivity* — ML boxes agreeing on the bottom-wire output at every input are equal — the crux, an induction on ML (LMHeadInj, 1 300 lines), the D box of a spine being read off by probing with Z and X on the bottom wire.

Relation reduction. The relations ract-sound uses are the paper's box relations, nine families of case tables quantified over the box labels (42 parametric relations against the paper's 66, differing only in grouping), derived (BR, 18 000 lines with its supporting calculi) from the *full* symplectic rule set of seventeen axioms: Selinger's c10–c15 [Selinger 2015] verbatim on two and three wires, and on one wire the orders of S , H and SH , $H^2S = SH^2$ and four multiplier *schemes* over units such as $M_x M_y = M_{xy}$. The *simplified* set derives all of these from relations about the single multiplier M_g of the primitive root (M -power $M_g^k = M_{g^k}$, semi-MS $M_g S = S^{g^2} M_g$, M_g past CZ on either wire, $H^2 = M_{-1}$; Appendix B), with c10–c15 and the orders of S and CZ : fifteen rules, eighteen with the structural ones; the identity on words is an isomorphism both ways (Simplified. Iso, 900 lines; the paper's Theorem 4.4). The odd prime shows only on one wire; the two- and three-wire rewriting is Selinger's. Completeness follows by by-normalization, surjectivity is "clear the bottom wire, recurse" (Theorem-LM, the paper's Lemma 3.7); and Theorem-Sp transports the presentation to the simplified rules along the identity isomorphism.

6 COMPOSING: PAULIS, THE SEMIDIRECT PRODUCT, AND SCALARS

$\overline{P}_n \cong (\mathbb{Z}_p \times \mathbb{Z}_p)^n$ is presented over X, Z on each wire by $X^p = Z^p = \varepsilon$ and $ZX = XZ$, with the phase space that $\text{Sp}(2n, \mathbb{Z}_p)$ acts on as its semantic group and normal form $X^a Z^b$ per wire. The semidirect product is the wreath example with $\text{Sp}(2n, \mathbb{Z}_p)$ for S_n : both factors' relations plus $h n h^{-1} = \text{conj } h n$, conj being the paper's Lemmas 2.22–2.23 as words. Its two conditions are proved by semantics: the conjugates of congruent words have equal denotations by symplectic soundness, hence are congruent by Pauli *completeness* (respects- Δ ; the paper's "Pauli complete" rules, Definition 4.7). Every premise being a theorem, the semidirect presentation of Pauli $\times$ Sp is unconditional.

Plumbing. The semidirect presentation has X, Z, H, S, CZ and $3+15+14$ relations (Pauli, symplectic, conjugation); Simplified-V1 has H, S, CZ and nineteen; Paper-V0 and Paper-V1 sixteen and fifteen (generator reduction: X, Z become words; relation reduction: Pauli and conjugation rules become theorems). *Semidirect to V1* is an isomorphism on *different* alphabets, f sending S to R and h sending S to $Z^{-1/2} \bullet S$, because V1 spells $M_x = R^x H R^{x^{-1}} H R^x H$ exactly; V1's nineteen rules add

(SH)³ = ε , $H^2SH^2S = SH^2SH^2$ and the two X -through- CZ rules to the simplified symplectic ones, and f-well-defined derives the three Pauli and fourteen conjugation rules (6 000 lines). *V1 to Paper-V0* is the identity on words: ten axioms are shared, *V0's* Lemmas (4 750 lines) derives c10–c15 from its own axioms. *V0 to the paper's fifteen* connects the routes: Figure 1 spells $M_x = SHS' x^{-1} x$ with $SHS' a b = Z^{(b-1)/2} X^{(1-a)/2} S^b H S^a H S^b H$ and semi-MR as $M_g S^{g^2} Z^{(g^2-g)/2} = S M_g$; Paper-V1 restates the four multiplier rules over SHS' and has fifteen rules, since under the S -spelling M_1 is (SH)³ and M -power at $k = 0$ gives $M_1 \approx \varepsilon$, whereas (RH)³ is a different word; the bridge $R^a H R^b H R^a H \approx SHS' a b$ (Shared.PauliBase) carries the multiplier rules across. Hence Theorem 2.1.

Scalars. The projective presentation is extended — add ω and $\omega^p = \varepsilon$, let ω commute with every gate, *correct* each rule by the scalar it picks up — by extension-presentation with a correction that is ε except on order- SH : with H normalised by δ_p exactly one rule fails on the nose, (SH)³ = ε picking up $\omega^{(p^2-1)/8}$ (residue $-1/8$), so it becomes (SH)³ = $\omega^{-1/8}$ (the multiplier rules hold exactly because R has no Pauli part). The semantic extension is the Weyl cocycle $c((P, S), (Q, _)) = \frac{1}{2} \text{sform}(P, \text{ap } S \ Q)$ on the natural semidirect product, whose cocycle law is symplectic invariance of sform ; the presented group's cocycle is pulled back from it, so the two agree by construction. On no hypothesis, presentation-exact gives, for every n , ($n \text{ Exact, } _ = _$) $\text{IsPresentationOf}(\text{Presented-group } n \text{ (exact-generator-data } n))$; it needs no normal forms, only that each rule's Weyl phase on the left equals the correction plus that on the right, nineteen $\mathbb{Z}_p$ equations. The extension is provably non-split.

7 DISCUSSION, RELATED WORK AND CONCLUSION

Everything above is checked by Agda 2.8 with standard library 2.4 under --safe , no postulates, no holes. Not formalised are the *glue facts* — that the constructed groups are the groups of unitaries generated by H, S, CZ modulo (or not) phases, the paper's Proposition 2.25 and Theorem 2.26 after Weil; no matrix appears in the development — the isomorphism of the two *total* central extensions, and the existence of primitive roots. The development is 106 388 lines in 329 files (Table 1): framework 15 000, qupit-specific 60 000, of which the symplectic factor is 35 000 and the composition step 23 000. *Lessons:* prove well-definedness by semantics; transport, do not re-prove; make the semantics compute. A language model helped write the Agda code and drafted this paper from the authors' outline.

Related work. Presentations of gate groups with hand-checked normal forms run from Selinger's n -qubit Clifford group [Selinger 2015] through [Amy et al. 2018, 2024; Bian 2023; Bian and Selinger 2021, 2023a,b; Greylyn 2014; Li et al. 2021; Makary et al. 2021] to the qutrit [Li et al. 2025] and qupit [Bian et al. 2026a] rules; the PROP-based programme [Blake 2026a,b,c; Clément 2026; Clément et al. 2024a,b, 2023], after Lafont [Lafont 2003], and the complete ZX calculi [Backens 2014; Booth and Carette 2022; Jeandel et al. 2018; Poór et al. 2023] are pen-and-paper too. Mechanised completeness is rare (Π as a free symmetric rig groupoid [Choudhury et al. 2022] and its square-root extension [Carette et al. 2024] are coherence arguments); verified compilers and logics [Chareton et al. 2021; Hietala et al. 2021; Lehmann et al. 2025; Li et al. 2020, 2022; Paykin et al. 2017; Peduri et al. 2025; Zhou et al. 2023] prove soundness of rewrites, never completeness; circuit optimisers [Kang and Oh 2023; Li et al. 2024; Paradis et al. 2024] search over rewrites; mathlib [The mathlib Community 2020, 2026], the AFP [Breitner 2010; Kharim et al. 2023] and MathComp [Gonthier et al. 2013] have no verified coset enumeration against a given concrete group.

Next are a verified matrix semantics closing the chain to $U_{p^n}(\mathbb{Z}[1/p, \omega])$, the qubit Clifford group [Selinger 2015], the qutrit rules [Li et al. 2025] as the case $p = 3$, and submission to the Agda standard library.

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A THE RULES, THE NORMAL BOXES, AND THE SIZE OF THE DEVELOPMENT

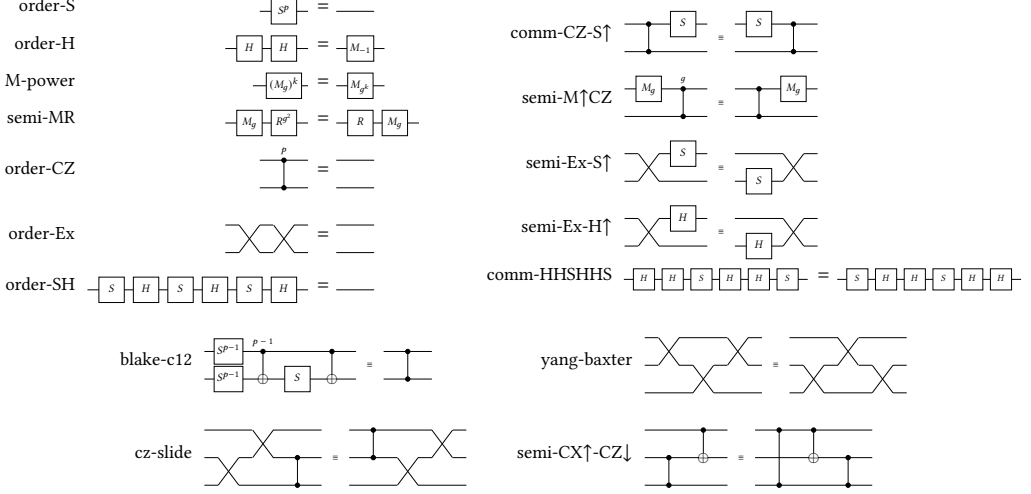


Fig. 1. The sixteen rules of $_CRel_==_$ (§2) as circuit equations, drawn from the Agda terms. Diagrams read in matrix-multiplication order, so a word $w \bullet v$ is drawn with v on the left. blake-c12 relates two circuits implementing controlled-Z; yang-baxter and cz-slide are the swap calculus. With the scalar restored (§6) only order-SH changes, to $(SH)^3 = \omega^{-1/8}$.

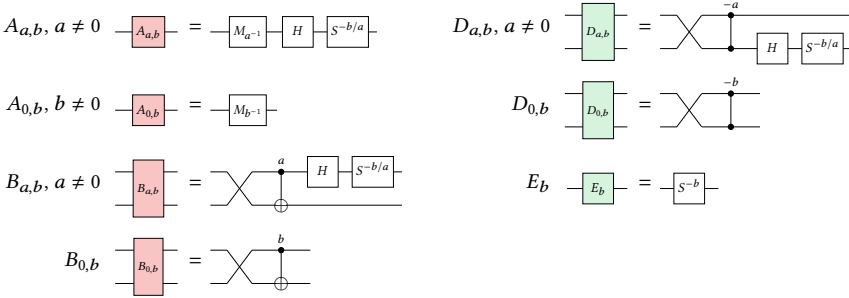


Fig. 2. The four kinds of normal box (the qubit paper’s Figure 6) and how each is spelled as a circuit by the section of the normal form (§5).

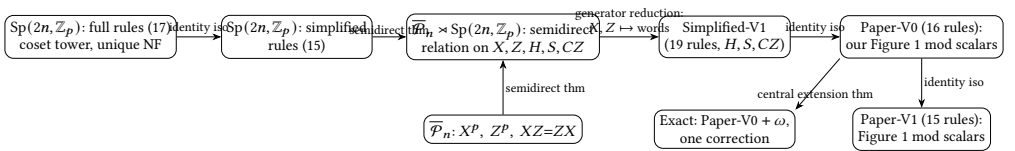


Fig. 3. The chain of presentations: every arrow is a verified presentation theorem or isomorphism of presentations; the semantic group is $\text{Sp}(2n, \mathbb{Z}_p)$ on the left, $\overline{\mathcal{P}}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p)$ from the semidirect level on, and a central extension of it by $\langle \omega \rangle$ at “Exact”.

Table 1. Size of the development (lines of Agda incl. comments, wc -l, September 2026).

Component	Files	Lines
Framework (Word, Presentation, Circuit, Normalization, ForStdlib)	54	14 848
Symplectic group (normalisation 8 352; box relations 18 161; soundness etc. 8 886)	118	35 399
Projective Pauli group	3	1 220
Projective Clifford: semidirect, plumbing, shared calculus	26	17 996
Clifford with scalars	11	5 029
Other examples in the tree (symmetric, cyclic, wreath, Clifford+T, qubit Clifford)	107	27 552
Not on the chain (Simplified-V2), index, notations	10	4 344
Total	329	106 388

B SELECTED AGDA DEFINITIONS

The normal-form datatype; the five hypotheses against which the library verifies a coset table (§4), in the generic Transfer module; the six one-qupit rules of the simplified symplectic rule set (§5), which with order-CZ, comm-CZ-S↓, comm-CZ-S↑ and selinger-c10–c15 make its fifteen; and the conjugation action of the semidirect presentation (§6), the paper’s Lemmas 2.22–2.23 as words. All three are quoted from the development, as is the doubly inductive normal-form datatype of §5, shown first.

```

ML ML : ℕ → Set
M 0 = T
M (1+ n) = Vec D n × E
L 0 = T
L (1+ n) = Vec B n × A
ML 0 = T
ML 1 = M 1 × L 1
ML m@(2+ n) = M m × L m
              ⊔ D × ML (1+ n)

NF : ℕ → Set
NF 0 = T
NF (1+ n) = NF n × ML (1+ n)

[] : ∀ {n} → NF n → Word (Gen n)
[] {0}    tt      = ε
[] {1+ n} (nf , ml) = [ nf ] ↑ • [ ml ]m1

(h=-1fgen : ∀ (x : X) → ([ x ]w , l) ~ ((ht) l (f x)))      - 1
(h=wd-ax : ∀ (c : C){u t : Word Y} → u ==2 t → ((ht) c u) ~ ((ht) c t)) - 2
(f=wd-ax : ∀ {w v} → w ==1 v → (fw) w ≈2 (fw) v)      - 3
([l]≈ε : [ l ] ≈2 ε)                                         - 4
(h=ract : ∀ c b → let (b' , c') = h c b in [ c ] • [ b ]w ≈2 (fw) b' • [ c' ]) - 5

order-S :   (1+ n) SRel,   S ^ p == ε
order-H :   (1+ n) SRel,   H ^ 2 == M.1
M-power : ∀ k → (1+ n) SRel, Mg ^ k == M (g ^ k)
semi-MS :   (1+ n) SRel,   Mg • S == S ^ (g * g) • Mg
semi-M↑CZ : (2+ n) SRel,   Mg ↑ • CZ == CZ ^ g • Mg ↑
semi-M↓CZ : (2+ n) SRel,   Mg ↓ • CZ == CZ ^ g • Mg ↓

conj : Sym.Gen n → XZ.Gen n → Word (XZ.Gen n)
conj Sym.H-gen X-gen   = Z
conj Sym.H-gen Z-gen   = X ^ p-1
conj Sym.S-gen X-gen   = X • Z
conj Sym.S-gen Z-gen   = Z
conj Sym.CZ-gen X-gen  = X • Z ↑
conj Sym.CZ-gen Z-gen  = Z
conj Sym.CZ-gen (X-gen ↑) = X ↑ • Z
conj Sym.CZ-gen (Z-gen ↑) = Z ↑
conj Sym.H-gen (xz ↑)   = [ xz ]w ↑
conj Sym.S-gen (xz ↑)   = [ xz ]w ↑
conj Sym.CZ-gen (xz ↑ ↑) = [ xz ]w ↑ ↑
conj (sym ↑) X-gen      = X
conj (sym ↑) Z-gen      = Z
conj (sym ↑) (xz ↑)     = conj sym xz ↑

```